1 Lab 01

1.1 Turn on an LED using an appropriate power source

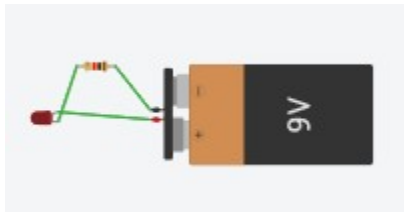

Figure 1: LED connected to a power source demonstrating basic LED operation.

1.2 Make an LED blink continuously using the Arduino Uno

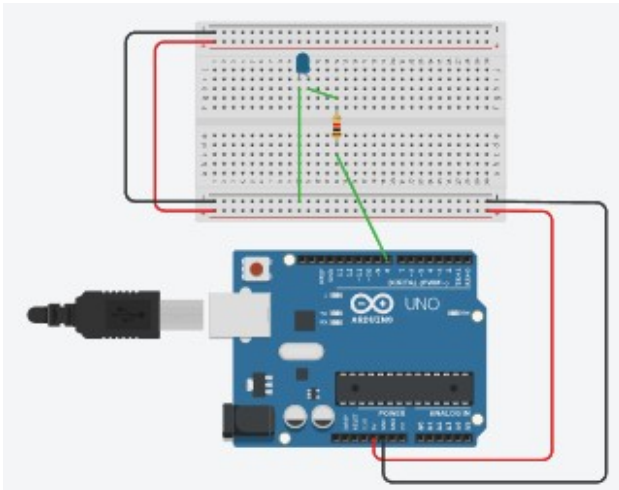

Figure 2: Arduino Uno circuit for continuously blinking an LED.

```
int led = 8;

void setup()
{
  pinMode(led, OUTPUT);
}

void loop()
{
  digitalWrite(led, HIGH);
  delay(1000);
  digitalWrite(led, LOW);
  delay(1000);
}
```

1.3 Implement a basic traffic light system using the Arduino Uno

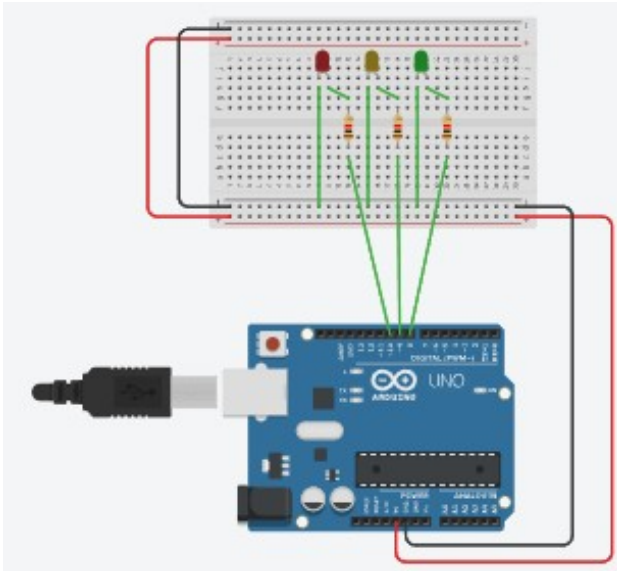

Figure 3: Arduino-based traffic light system using red, yellow, and green LEDs.

```
int red = 10;
int yellow = 9;
int green = 8;

void setup()
{
  pinMode(red, OUTPUT);
  pinMode(yellow, OUTPUT);
  pinMode(green, OUTPUT);
}

void loop()
{
  digitalWrite(red, HIGH);
  digitalWrite(yellow, LOW);
  digitalWrite(green, LOW);
  delay(2000);

  digitalWrite(red, LOW);
  digitalWrite(yellow, HIGH);
  digitalWrite(green, LOW);
  delay(2000);

  digitalWrite(red, LOW);
  digitalWrite(yellow, LOW);
  digitalWrite(green, HIGH);
  delay(2000);
}
```

2 Lab 02

2.1 Turn on an LED using a Push Button

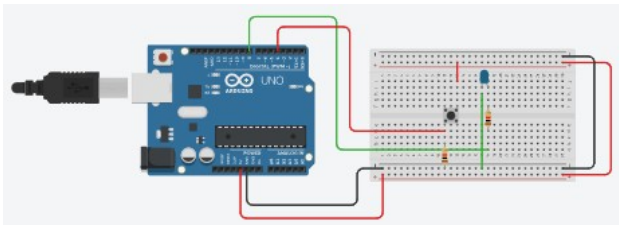

Figure 4: Push button controlling an LED using Arduino Uno.

```
int led = 8;
int button = 4;

void setup()
{
  pinMode(led, OUTPUT);
  pinMode(button, INPUT);
}

void loop()
{
  int buttonState = digitalRead(button);

  if(buttonState == HIGH)
  {
    digitalWrite(led, HIGH);
  }
  else
  {
    digitalWrite(led, LOW);
  }
}
```

2.2 Make an LED state toggle each time the button is pressed

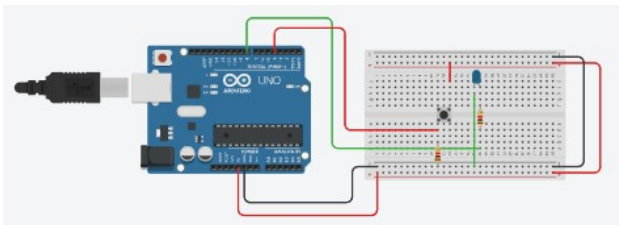

Figure 5: Button press toggling the LED state between ON and OFF.

```
int led = 8;
int button = 4;
int ledState = LOW;
int lastButtonState = LOW;

void setup()
{
  pinMode(led, OUTPUT);
  pinMode(button, INPUT);
}

void loop()
{
  int buttonState = digitalRead(button);

  if (buttonState == HIGH && lastButtonState == LOW)
  {
    ledState = !ledState;
    digitalWrite(led, ledState);
  }

  lastButtonState = buttonState;
}
```

2.3 Button-controlled LED blink

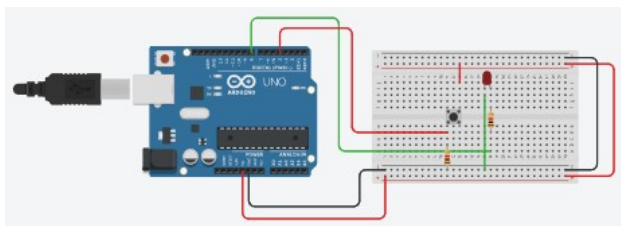

Figure 6: Button-controlled LED blinking system using Arduino.

```
int led = 8;
int button = 4;
int ledState = HIGH;
int lastButtonState = LOW;
bool blinking = false;
unsigned long previousMillis = 0;
const long interval = 500;

void setup()
{
  pinMode(led, OUTPUT);
  pinMode(button, INPUT);
  digitalWrite(led, ledState);
}

void loop()
{
  int buttonState = digitalRead(button);

  if (buttonState == HIGH && lastButtonState == LOW)
  {
```

```
    blinking = !blinking;
    if (!blinking)
    {
      ledState = HIGH;
      digitalWrite(led, ledState);
    }
  }

  lastButtonState = buttonState;

  if (blinking)
  {
    unsigned long currentMillis = millis();

    if (currentMillis - previousMillis >= interval)
    {
      previousMillis = currentMillis;
      ledState = !ledState;
      digitalWrite(led, ledState);
    }
  }
}
```

3 Lab 03

3.1 Make an LED state toggle each time the button is pressed using an interrupt

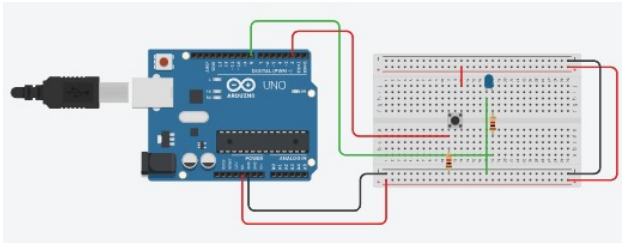

Figure 7: Interrupt-based button control for toggling LED state.

```
int led = 8;
int button = 2;

bool ledState = false;

void setup()
{
  pinMode(led, OUTPUT);
  pinMode(button, INPUT);
  Serial.begin(9600);
  attachInterrupt(digitalPinToInterrupt(button), ledToggle, FALLING);
}

```

```
void loop()
{
  digitalWrite(led, ledState);
}

void ledToggle()
{
  ledState = !ledState;
}
```

3.2 Control LED brightness using a Potentiometer

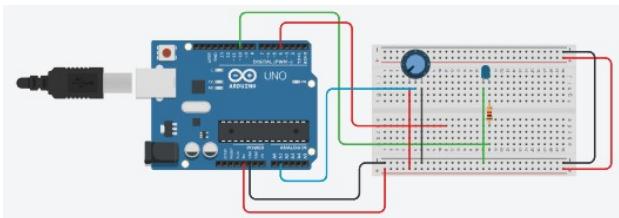

Figure 8: LED brightness control using a potentiometer and PWM signal.

```
int led = 10;
int potn = A1;

void setup()
{
  pinMode(led, OUTPUT);
  Serial.begin(9600);
}

void loop()
{
  int potnValue = analogRead(potn);
  int brightness = map(potnValue, 0, 1023, 0, 255);

  analogWrite(led, brightness);
  Serial.println(brightness);
}
```

3.3 Turn on another LED based on the medium brightness (180) of an LED

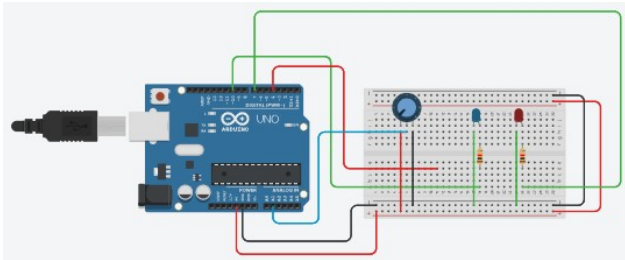

Figure 9: Second LED turning ON when the first LED brightness exceeds a threshold value.

```
int led = 10;
int potn = A1;
int led2 = 7;

void setup()
{
  pinMode(led, OUTPUT);
  pinMode(led2, OUTPUT);
  Serial.begin(9600);
}

void loop()
{
  int potnValue = analogRead(potn);
  int brightness = map(potnValue, 0, 1023, 0, 255);

  analogWrite(led, brightness);
  Serial.println(brightness);

  if (brightness >= 180)
  {
    digitalWrite(led2, HIGH);
  }
  else
  {
    digitalWrite(led2, LOW);
  }
}
```

4 Lab 04

4.1 Build a gas detector and alert system with a buzzer

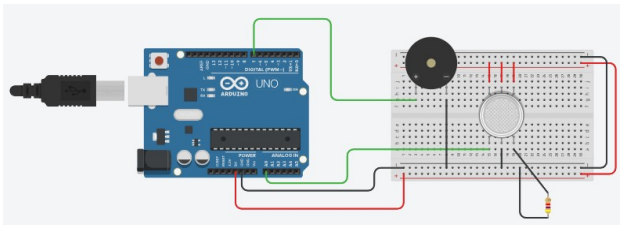

Figure 10: Gas detection system with buzzer alarm using a gas sensor.

```
int piezo = 7;
int gesensor = A0;

void setup()
{
  pinMode(piezo, OUTPUT);
  Serial.begin(9600);
}

void loop()
{
  int val = analogRead(A0);
  Serial.println(val);

  if(val > 350)
  {
    digitalWrite(piezo, HIGH);
  }
  else
  {
    digitalWrite(piezo, LOW);
  }
}
```

4.2 Turn on an LED based on a daylight sensor

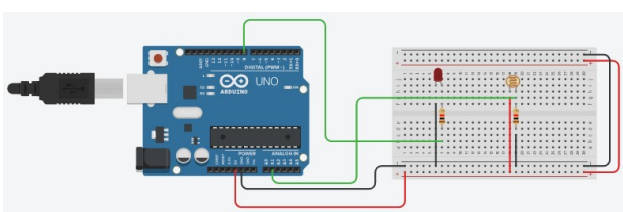

Figure 11: LED controlled by a daylight sensor using a photoresistor.

```
int photors = A1;
int led = 8;

void setup()
{
  pinMode(led, OUTPUT);
  Serial.begin(9600);
}

void loop()
{
  int val = analogRead(photors);
  Serial.println(val);

  if(val < 550)
  {
    digitalWrite(led, HIGH);
  }
  else
  {
    digitalWrite(led, LOW);
  }
}
```